

Luca Luigi Alberici

London | luca99alberici@gmail.com | +44 7895 818 462 | luca-luigi-alberici.github.io | [LinkedIn](#)

SUMMARY

Doctoral Candidate in Finance at Bayes Business School specializing in Financial Engineering and Mathematical Finance. My research focuses on statistical learning, machine learning, and quantitative risk management, with applications to asset pricing, forecasting, and financial markets. I develop high-dimensional models using Bayesian inference, latent factor structures, and EM-based estimation techniques. My current work also investigates liquidity provision risk in decentralized finance and the use of advanced statistical methods for financial decision-making under uncertainty.

RESEARCH

Working papers.

L.L. Alberici, L. Ballotta, G. Fusai (2026). *Disciplined Learning: A GenAI Framework for Arbitrage-Free Term Structures*.

L.L. Alberici, L. Ballotta, G. Fusai (2026). *Estimation of a Hierarchical Normal-Inverse Gaussian Factor Model via the EM Algorithm*.

L.L. Alberici, A. Papanicolaou, L. Schoenleber (2025). *Implied Impermanent Loss for Concentrated Liquidity*.

Work in progress.

L.L. Alberici, F. Centrone, A. Goia (2026). *Technical Efficiency: A Regularization-Based Framework*.

CONFERENCES

Implied Impermanent Loss for Concentrated Liquidity

- **2026:** *XIII Bachelier World Congress*, Bachelier Finance Society, Bologna; *8th QMUL Economics and Finance Workshop*, London; *3rd QRFE Workshop on Quantitative Finance*, Durham; *8th Blockchain International Scientific Conference (ISC2026)*, London; *Decentralized Finance & Crypto Workshop*, Pisa; *2026 Joint Mathematics Meetings (JMM)**, Washington DC.
- **2025:** *Seminars in Statistics**, Turin; *Algo-trading & DeFi Conference*, Milan; *Stanford Applied Math Seminar**, Stanford.

Estimation of a Hierarchical Normal-Inverse Gaussian Factor Model via the EM Algorithm

- **2026:** *International Conference on Computational Finance*, University of Oxford, Oxford; *XIII Bachelier World Congress*, Bachelier Finance Society, Bologna; *PhD Finance and Accounting Research Day*, Bayes Business School, London; *Seminari Matematici Statistici*, Novara; *FAM*, London.

The symbol * indicates conferences attended by the co-authors.

Organizer of Workshop

- **2026:** *Financial Engineering Workshop*, Bayes Business School, London.

EDUCATION

Bayes Business School, PhD in Finance, *London (UK)*

Sept 2025 – Sept 2028

- Doctor of Philosophy (DPhil) in Finance.
- Awarded full-tuition fee waiver based on academic distinction.
- Co-leading the foundation of the PhD's Alumni Association.

Bayes Business School, Master of Philosophy (MPhil) in Finance, *London (UK)*

Sept 2024 – Sept 2025

- Core Modules: Asset Pricing, Financial Econometrics, Banking, Market Microstructure, Machine Learning.
- Awarded full-tuition fee waiver based on academic distinction.

Collegio Carlo Alberto, MSc in Finance, Insurance & Risk Management, *Turin (IT)*

Sept 2023 – July 2024

- Second Level MSc, specialization in Quantitative Finance.
- Main courses: Asset Pricing, Asset Allocation, Derivatives, Fixed Income, Machine Learning, Optimization.

- Merit-based partial tuition fee waiver of 90%.
- Co-leading the foundation of the Master's Alumni Association.

Università del Piemonte Orientale, MSc in Management and Finance, *Novara (IT)* Sept 2021 – Nov 2023

- Final grade: 110/110 cum laude with Academic Mention for the career.
- Student representative.

Università del Piemonte Orientale, BSc in Business Administration, *Novara (IT)* Sept 2018 – July 2021

Final grade: 110/110 cum laude.

PhD Courses & Schools

- **Scuola Normale Superiore**, Pisa (IT) – *Mathematical Models for Quantitative Finance: Market Microstructure, Networks and Systemic Risk* (Prof. Lillo), Feb 2026
- **Università Ca' Foscari**, Venice (IT) – *Doctoral Colloquium on Risk Analytics: Modern Risk Measurement* (Proff. Li Calzi & Corazza), Mar 2025
- **Politecnico di Milano**, Milan (IT) – *Optimal Stochastic Control Methods in Mathematical Finance* (Prof. Marazzina), Feb 2025
- **Università degli Studi Roma Tre**, Rome (IT) – *AI Algorithms, Graph Theory, Stochastic Dynamic Optimization* (Prof. Cesarone), May 2024

Exchange & International Experience

- **Bloomberg Training Program**, Stevens Institute of Technology, Hoboken (US) – free mover funded by a scholarship (€1,500.00), Jun 2022
- **Functional Data Analysis**, Universitat Politècnica de Catalunya, Barcelona (ES) – R programming in Functional Data Analysis (Time Series for Stock Market), free mover funded by a scholarship, Jun 2021

TEACHING EXPERIENCE

Module Leader in Data Analytics (*upcoming*), Bayes Business School, *London (UK)* Sept 2026
Tools: Excel.

Teaching Certificate: Advance Higher Education (UK), Bayes Business School, *London (UK)* March 2026
Qualification for Teaching in UK

GTA in Financial Econometrics, Imperial College Business School, *London (UK)* Sept 2025 – April 2026
Module Leader: Dr. Roald Versteeg
Topics: regression & OLS diagnostics, ARMA modelling & forecasting, unit roots, rolling-window factor models, bootstrap inference, cointegration & pairs trading, volatility modelling, ML in financial econometrics. Tools: R, Python.

GTA in Financial Econometrics, Bayes Business School, *London (UK)* Sept 2025 – April 2026
Module Leader: Dr. Parastoo Mousavi
Topics: EViews & stock market data, (log-)returns & descriptive statistics, OLS regression & residual diagnostics, CAPM & factor models, cross-sectional regression, return predictability. Tools: EViews.

GTA in Applied Business Intelligence, Bayes Business School, *London (UK)* Sept 2025 – December 2025
Module Leader: Dr. Parastoo Mousavi
Topics: Tableau fundamentals, charts & dashboards, exploratory analysis (library, healthcare, transport data), geographic & spatial visualisation, time series dashboards. Tools: Tableau.

GTA in Financial and Business Analytics, Bayes Business School, *London (UK)* Sept 2025 – December 2025
Module Leader: Dr. Parastoo Mousavi
Topics: applied econometrics for financial data, time series & volatility modelling, portfolio construction & performance evaluation, cross-sectional regression & factor models, forecasting. Tools: EViews.

GTA in Pre-Study for Mathematics, Statistics and Python, Bayes Business School, *London (UK)* Sept 2025 – December 2025
Module Leader: Dr. Elisabetta Pellini
Topics: algebra & calculus refresher, linear algebra, probability theory, descriptive/inferential statistics, hypothesis testing, intro to Python for data analysis. Tools: Python.

GTA in Mathematics and Statistics, Università del Piemonte Orientale, Novara (IT) Sept 2021 – June 2023
Funded by a scholarship (€4,000.00)
Topics: limits, continuity & derivatives, integral calculus, sequences & series, descriptive/inferential statistics, probability distributions, hypothesis testing. Tools: R.

PROFESSIONAL EXPERIENCE

Institute and Faculty of Actuaries (IFoA), UK June 2026 – present
Member, IFoA General Insurance – Machine Learning in Reserving Working Party
Investigating applications of machine learning in general insurance reserving.
Contribute to industry research, methodological development, and knowledge-sharing initiatives at the intersection of actuarial science, finance, and machine learning.

Università del Piemonte Orientale Sept 2021 – June 2023
Team leader for library tutors

Leo Club Biraga (voluntary work) June 2018 – July 2024

- Treasurer's office.
- Environmental manager.

Supervision in Italian Elections June 2017 – present

PROJECTS

Perfect Withdrawal Rate Analysis Dec 2025
Developed a Python framework to compute and simulate the Perfect Withdrawal Rate over a fixed horizon without depleting a portfolio below a target terminal value)

Data Envelopment Analysis in Portfolio Theory: an Empirical Application Sept 2022 – Nov 2023
New approach to stock selection (ranking, efficiency score, optimization)

- Ranking of companies given risk and performance measures of a stock market (FTSE MIB) to better select the companies in which to invest.
- Research award, "Ottavio Signorelli" scholarship (€2,000.00).

Karush–Kuhn–Tucker Theorem: an Empirical Application in Portfolio Theory Nov 2020 – July 2021
Different approach to portfolio optimization (minimum threshold of expected return)

- Study of the Karush-Kuhn-Tucker theorem. Comparison with the original solution of Markowitz problem.
- Research award, "Ottavio Signorelli" scholarship (€2,000.00).

Machine Learning Specialization June 2024 – Sept 2024
DeepLearning.AI, Stanford University

- Unsupervised Learning, Recommenders, Reinforcement Learning, Supervised Machine Learning: Regression and Classification, Advanced Learning Algorithms.

TECHNICAL SKILLS AND INTERESTS

Languages: Italian (native), English (fluent), French (basic)

Developer Tools: Python, MATLAB, R, EViews, SQL, High-Performance-Computing (HPC), OxMetrics, Bloomberg, LSEG, VBA, HTML, LaTeX, Tableau, Office (Word, Excel, Power Point)

Cloud: Personal GitHub for Projects (i.e. web-application in Streamlit and Excel365)

Soft Skills: team working, adaptability, problem solving

Hobbies: running (Cancer Research UK's Race for Life), hiking, amateur culinary competitions